

HW# 3

4.24

Suppose that the bandlimited differentiator of Example 4.4 has input $x_c(t) = \cos(\Omega_0 t)$ with $\Omega_0 < \pi/T$. In this problem, we wish to verify that the continuous-time signal reconstructed from the output of the bandlimited differentiator is indeed the derivative of $x_c(t)$.

- (a) The sampled input will be $x[n] = \cos(\omega_0 n)$, where $\omega_0 = \Omega_0 T < \pi$. Determine an expression for $X(e^{j\omega})$ that is valid for $|\omega| \leq \pi$.
- (b) Now use Eq. (4.46) to determine the DTFT of $Y(e^{j\omega})$, the output of the discrete-time system.
- (c) From Eq. (4.32) determine $Y_r(j\Omega)$, the continuous-time Fourier transform of the output of the D/C converter.
- (d) Use the result of (c) to show that

$$y_r(t) = -\Omega_0 \sin(\Omega_0 t) = \frac{d}{dt} [x_c(t)].$$

Remarks:

$$(1) Y_r(j\Omega) = H_r(j\Omega)Y(e^{j\omega})|_{\omega=\Omega T} = H_r(j\Omega)Y(e^{j\Omega T}) = \begin{cases} TY(e^{j\Omega T}), & |\Omega| < \frac{\pi}{T} \\ 0, & \text{otherwise} \end{cases} \quad (4.32)$$

$$(2) H(e^{j\omega}) = j\frac{\omega}{T}, \quad |\omega| < \pi. \quad (4.46)$$

5.8

A causal LTI system is described by the difference equation

$$y[n] = \frac{3}{2}y[n-1] + y[n-2] + x[n-1].$$

- (a) Determine the system function $H(z) = Y(z)/X(z)$ for this system. Plot the poles and zeros of $H(z)$, and indicate the ROC.
- (b) Determine the impulse response of the system.
- (c) You should have found the system to be unstable. Determine a stable (noncausal) impulse response that satisfies the difference equation.

5.12

A discrete-time causal LTI system has the system function

$$H(z) = \frac{(1 + 0.2z^{-1})(1 - 9z^{-2})}{(1 + 0.81z^{-2})}.$$

- (a) Is the system stable?
- (b) Determine expressions for a minimum-phase system $H_1(z)$ and an all-pass system $H_{ap}(z)$ such that

$$H(z) = H_1(z)H_{ap}(z).$$

5.15

Consider the class of discrete-time filters whose frequency response has the form

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{-j\alpha\omega},$$

where $|H(e^{j\omega})|$ is a real and nonnegative function of ω and α is a real constant. As discussed in Section 5.7.1, this class of filters is referred to as *linear-phase* filters.

Consider also the class of discrete-time filters whose frequency response has the form

$$H(e^{j\omega}) = A(e^{j\omega})e^{-j\alpha\omega + j\beta},$$

where $A(e^{j\omega})$ is a real function of ω , α is a real constant, and β is a real constant. As discussed in Section 5.7.2, filters in this class are referred to as *generalized linear-phase* filters.

For each of the filters in Figure P5.15, determine whether it is a generalized linear-phase filter. If it is, then find $A(e^{j\omega})$, α , and β . In addition, for each filter you determine to be a generalized linear-phase filter, indicate whether it also meets the more stringent criterion for being a linear-phase filter.

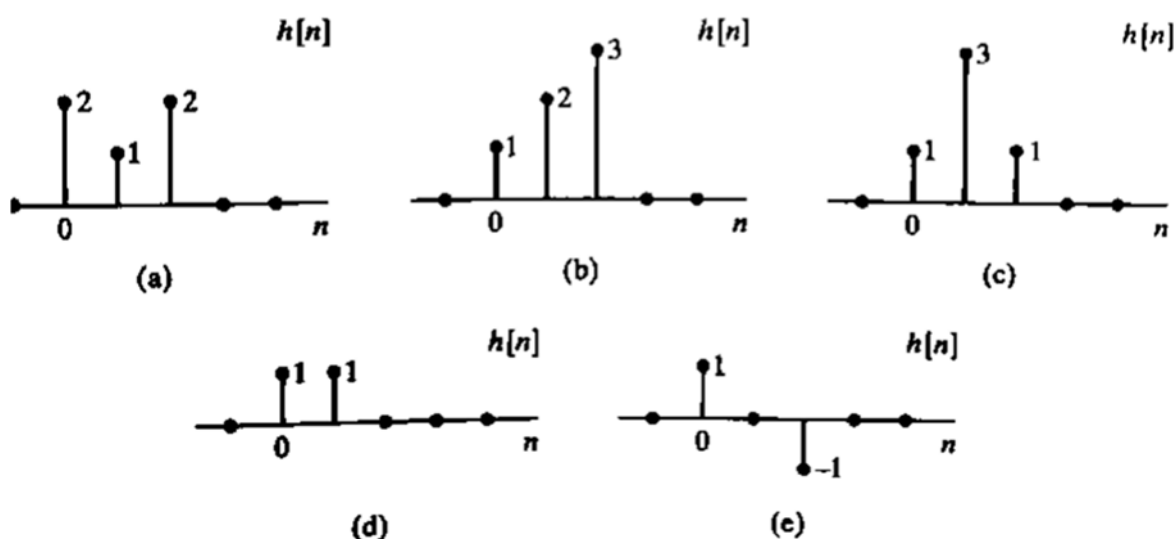


Figure P5.15

Matlab:

Plot the frequency response and group delay of

$$H(z) = \frac{0.001836(1+z^{-1})^4}{(1-1.5548z^{-1}+0.6493z^{-2})(1-1.4996z^{-1}+0.8482z^{-2})}$$